MIL-T-7748E
AMENDMENT 2
29 November 1978
SUPERSEDING
AMENDMENT 1
21 FEBRUARY 1973

MILITARY SPECIFICATION

TRANSMITTERS, PRESSURE, SYNCHRO, AIRCRAFT 320 DEGREE MOVEMENT, HIGH TEMPERATURE (177°C)

This amendment forms a part of Military Specification MIL-T-7748E dated 31 March 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 1

1.2 Make the two following changes:

a. Under the first column, add the four classifications in the blanks between MS28131-8T and MS28131-9T: MS28131-19T, MS28131-20T, MS28131-21T, and MS28131-22T.

b. Delete second column removing MS28131-20D through MS28131-32 from the classification.

Bottom of page add the following:

> Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, ESSD, (Code 93), Lakehurst, New Jersey 08733 by using the self-addressed Standard Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Page 2

2.1 Make the following two changes:

a. Delete MIL-L-5020 Liquid, Compass, Aircraft

b. Add MIL-L-7808 Lubricating Oil, Aircraft Turbine Engine, Synthetic Base

FSC 6620

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Page 5

3.4 Delete last sentence and columns and substitute the following:

"The inside diameter of the pressure tubing shall be between 0.031 and 0.046 inches."

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3.4.2.2 Delete and substitute the following:

"3.4.2.2 Filling medium: The case shall be filled with a suitable fluid to provide viscous damping of vibration and aircraft pump pulsations. Allowance shall be made for expansion of the damping fluid at high temperature by partially filling the case with helium gas. The helium gas shall contain not more than 0.006 milligrams of water vapor per liter (dew point -65°C) at the filling pressure. The absolute pressure of the helium in the case shall be 1 ± 0.1 atmosphere."

Page 7

3.4.4.1 Delete and substitute the following:

"3.4.4.1 All gage pressure transmitters of 150 psi or greater range shall have a case pressure release constructed so that if the sensor connected to the "P" port should leak, the case pressure would be relieved and the fluid discharged through the vent port. The port shall be marked 'V & CPR'."

3.4.4.1 Delete first sentence and substitute:

"The blow-out device for releasing case pressure shall discharge through a 5/16-24 connection conforming to MS33656 except the thread shall be as specified on MS28131."

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3.5.1 Delete and substitute the following:

"3.5.1 Primary input power requirements: Transmitters through the 500 psi range shall operate on ac power (single phase), 26v ± 2.0v, 400 Hz ± 5 percent, 5.20 va maximum and transmitters of 1000 psi and greater shall operate on ac power (single phase), 26v ± 2.0v, 400 Hz ± 5 percent, 13.00 va maximum."

3.5.5 Add the following sentence:

"Internal wiring shall not be spliced."

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4.4.2.1.1 Delete "(a) Damping (4.6.9)" and substitute "(a) (Blank)"

4.4.3 Delete and substitute the following:

"4.4.3 Reliability assurance tests: Reliability assurance tests shall be conducted in accordance with MIL-STD-781; first article (preproduction) qualification test and production acceptance (sampling) (for Army and Navy contracts only). Equipments selected for the reliability assurance tests shall first have passed the individual tests."

4.4.3.1 Delete and substitute the following:

"4.4.3.1 Reliability first article (preproduction) qualification test: Prior to Qualified Products Listing (QPL) (see 3.1) a minimum of three equipments shall be tested as specified in MIL-STD-781. Reliability prediction, a reliability test plan document, and a burn-in test are not required. The thermal survey report and vibration survey report may be incorporated as appendices to the reliability test procedure. The reliability test procedure shall include both qualification and acceptance test procedures. The maximum number of equipments to be used shall be seven. For both the qualification and acceptance tests the temperature shall be cycled between -54°C and 155°C which shall be time to stabilize at low temperature followed by time to stabilize plus two hours soak at high temperature. Vibration shall be sinusodial 10g ± 10 percent peak acceleration value at any nonresonant frequency between 180 and 240 Hz, measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. Input voltage shall be 26v +5, -2 percent. The accept/reject criteria of Test Plan VIC shall be used."

4.4.3.2 Delete and substitute the following:

"4.4.3.2 Production reliability acceptance test (sampling): Samples of the production equipment shall be tested as specified in MIL-STD-781. The test conditions specified in 4.4.3.1 shall apply. The accept/reject criteria of Test Plan VIIIC of MIL-STD-781 shall be used."

4.4.3.2.1 Delete and substitute the following:

"4.4.3.2.1 Procedure for production reliability acceptance test (sampling): The production reliability acceptance test (sampling) shall be conducted at least once on each lot. (The equipments constructed during a three month period or less shall constitute a lot.) The sample size shall be selected, when possible, so that the anticipated test period to an

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accept decision will not exceed 30 days. The test program shall begin with the first lot. The samples shall be tested until an accept or reject decision is reached. The procuring activity reserves the right to stop acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment quality control, etc."

Add a new paragraph:

"4.5.5 Test fluid: The transmitters shall be subjected to the inspection methods of 4.4.3 and 4.6 using aircraft turbine oil in accordance with MIL-T-7808."

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4.6.9 Delete and substitute the following: "4.6.9 - (Blank)"

4.6.10 Delete the first sentence and substitute the following:

"Gage pressure transmitters, with part numbers MS28131-2T, -4T, and -6T shall have a vacuum of 0.5 ± 0.5 psi absolute pressure applied to the "P" port for 5 minutes with the "V" port open to the atmosphere."

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4.6.14 Delete last sentence.

4.6.16 Delete last sentence.

Page 23

4.6.20 Delete text and substitute the following:

"The transmitter shall be tested and meet the requirements of MIL-T-5350 except that the open secondary current of the transmitters through the 500 psi range shall not exceed 200 milliamperes, and the input power comsumption shall not exceed 5.20 voltamperes; transmitters of 1000 psi and greater range shall not exceed 500 milliamperes with a maximum input power consumption of 13.00 voltamperes."

Page 26

6.2 (e) Delete and substitute the following:

"(e) Reliability assurance (see 4.4.3) requiring:
(1) Reports Reliability Test and Demonstration (DI-R-7034)
(2) Procedures, Reliability Test (DI-R-7035)
(3) Report, Thermal Survey (DI-R-7036)

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(4) Report, Vibration Survey (DI-R-7037)
(5) Plan, Corrective Action (DI-R-7038)
(6) Report, Failed Item Analysis (DI-R-7039)
(7) Report, Failure Summary and Analysis (DI-R-7041)

6.3.4 Delete text and substitute the following:

"'Good as new' shall mean transmitters operated for less than 400 hours."

Custodians:
Navy - AS
Air Force - 11

Reviewing Activities:

Army - AV
Air Force - 99

User Activity:
Navy - CG

Preparing Activity:
Navy - AS
(Project No. 6620-0310)

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